Akka vs. AWS Bedrock (AgentCore)

A comparison for teams building agentic AI

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AWS Bedrock AgentCore is a set of separate AWS services you wire and operate yourself — not an integrated, governed platform. AgentCore gives you modular building blocks (including Runtime, Memory, Gateway, Identity, Code Interpreter, Browser, Observability, Policy, Evaluations, Web Search) that you assemble on top of Bedrock model access, Guardrails, and Knowledge Bases — each billed and operated separately, all running inside AWS. The Akka Agentic AI Platform delivers orchestration, agents, memory, streaming, APIs, observability, and governance as one runtime that guarantees resilience and scalability, with a six-nines SLA Akka owns, deployable on any cloud.

99.9%	99.9999%	Separate	Oct 2025
BEDROCK SLA	AKKA SLA	AGENTCORE SERVICES TO WIRE	AGENTCORE GA

DIMENSION	AWS BEDROCK AGENTCORE	AKKA
What it is	A set of modular AWS services for building and hosting agents	The Akka Agentic AI Platform — a platform that guarantees resilience and scalability
Scope	AgentCore services assembled on Bedrock model access, Guardrails, and Knowledge Bases — the customer wires and operates the seams	Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime
Availability SLA	99.9% monthly for the Bedrock APIs, measured by request errors; service credits only	99.9999% — entire platform, backed by indemnities
RTO / RPO	Not published for the agent layer; inherits the AWS regional posture the customer architects	Sub-1-minute RTO; zero-byte RPO; active-active across regions
Deployment	AWS only — IAM, VPC/PrivateLink, CloudFormation, CloudWatch	Any cloud (AWS/Azure/GCP VPC), own Kubernetes, on-prem, sovereign cloud
Memory	AgentCore Memory — a separate, separately-billed service	Durable in-memory, 4ms reads / sub-10ms writes, built in
Governance / EU AI Act	AgentCore Policy blocks tool calls at the Gateway, with Guardrails in Policy and human approval through temporal policies; no mapping to a named regulation	Aspect-woven runtime enforcement + pre-production classification against 241 regs / 1,718 controls
Cost model	Consumption-metered across the AgentCore services (per-vCPU-hour, per-request, per-token) plus separately billed model inference	Shared compute; up to 90% lower infrastructure for the same workload, fixed annual fee
Maturity	GA October 2025; Policy and Evaluations GA March 2026; Guardrails in Policy GA June 2026 — capabilities still landing	since 2009; 100,000+ production deployments; 52 banks
Certifications	Bedrock: ISO, SOC, CSA STAR L2, GDPR, FedRAMP High, HIPAA-eligible	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, EU AI Act, NIST AI RMF)

AgentCore Is a Set of Services to Assemble; Akka Is One Platform

AWS Bedrock AgentCore is a collection of separate AWS services the customer integrates and operates, not an integrated platform. At general availability in October 2025 it has added services; the pricing page now lists services including Runtime, Memory, Gateway, Identity, Code Interpreter, Browser, Observability, Policy, Evaluations, and Web Search — and an agent of substance also draws on

Bedrock model access, Bedrock Guardrails, and Knowledge Bases. Each is provisioned, secured, billed, and operated on its own; the customer owns every seam between them.

Capability	AWS Bedrock AgentCore	Akka
Agent hosting / runtime	AgentCore Runtime (separate service)	Built in
Durable memory	AgentCore Memory (separate, separately billed)	Built in, 4ms / sub-10ms
Tool / API gateway	AgentCore Gateway (separate service)	Built in
Identity & authorization	AgentCore Identity (separate service)	Built in
Observability	AgentCore Observability on CloudWatch (separate)	Built in
Model access	Amazon Bedrock (separately billed per token)	Bring any model
Real-time streaming	Not a native AgentCore service	Built in, backpressured, petabyte-scale
Governance / policy enforcement	AgentCore Policy at the Gateway, with Guardrails in Policy; agents on AgentCore only	Inline, runtime-embedded

Akka delivers all of this on one runtime with one operational model. There are no seams to integrate, no per-service SLAs to reconcile, and no cross-service failure modes the customer inherits.

Availability and Disaster Recovery

The agent layer has no dedicated availability SLA. Amazon Bedrock publishes a 99.9% monthly uptime commitment, measured as the percentage of Bedrock API requests that do not fail with errors, calculated per region, with service credits as the sole remedy. AgentCore inherits the regional, single-cloud posture the customer architects on top of AWS — the customer owns multi-region failover, RTO, and RPO for the assembled system.

Metric	AWS Bedrock AgentCore	Akka
Availability SLA	99.9% monthly (Bedrock API errors)	99.9999%
Allowed downtime / year	~8.8 hours	~31 seconds
RTO	Customer-architected	Sub-1 minute
RPO	Customer-architected	Zero byte
SLA scope	The Bedrock model APIs	The entire platform
Remedy	Service credits	Contractual indemnities

99.9% per region versus 99.9999% is the difference between roughly 8.8 hours and 31 seconds of allowed downtime a year. The Akka SLA covers the whole running system — orchestration, agents, memory, streaming, APIs, and governance — and Akka owns it with 24/7 SRE.

Continuous AI Intelligence

Akka Optimize keeps improving the AI after it ships. It runs an evaluation loop on your own production traffic, grading live interactions and training smaller specialized models on your proprietary data, so the system keeps getting better as it runs. The loop delivers three results that compound over time:

- **Model leverage** — route each request to the best model from any vendor under your policies, and reserve frontier models for the requests that need them.
- **Data sovereignty** — the specialized models are trained on your proprietary data, owned by you, and run inside your own environment.
- **Cost governance** — full visibility and control of AI spend, with savings that compound as the loop keeps running.

The evaluations run continuously inside Akka and grade traffic from agents wherever they run, in Akka or in third-party harnesses.

Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume, not a list-price comparison. AgentCore meters its services independently — Runtime, Memory, Gateway, Identity, Code Interpreter, Browser, and Observability, plus Evaluations and Policy — on per-vCPU-hour, per-request, and per-token units, on top of standard Bedrock model token spend. Every layer the agent touches is a separate consumption meter that moves with load.

Akka runs all of it on one shared-compute runtime. The efficiency comes from actor concurrency (~10T tokens/core/year vs. ~2T; ~80% less compute than Python frameworks; Manulife: up to 300% more concurrency, 30–50% faster processing after porting from Python), shared compute, and micro-checkpointing that minimizes retries. The spend is a fixed annual fee — not a stack of usage meters that scale with traffic.

Governance and the EU AI Act

AgentCore Policy checks each tool call that passes through an AgentCore Gateway and denies calls that break a rule before they run; Policy reached GA in March 2026. Since June 2026, Bedrock Guardrails screen the same calls for prompt injection and sensitive data. Since August 2026, temporal policies can require human approval before a privileged action. The controls cover agents whose traffic runs through an AgentCore Gateway on AWS. AWS maps them to no named regulation. Audit Manager's generative AI framework (110 controls) and Security Hub's AI Security Best Practices standard (31 controls) check AWS recommendations. There is no built-in classification against the EU AI Act.

The EU AI Act Penalties Are Enforceable Now

What a violation costs

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover
Incorrect information (Art. 99)	EUR 7.5M or 1.5% global turnover

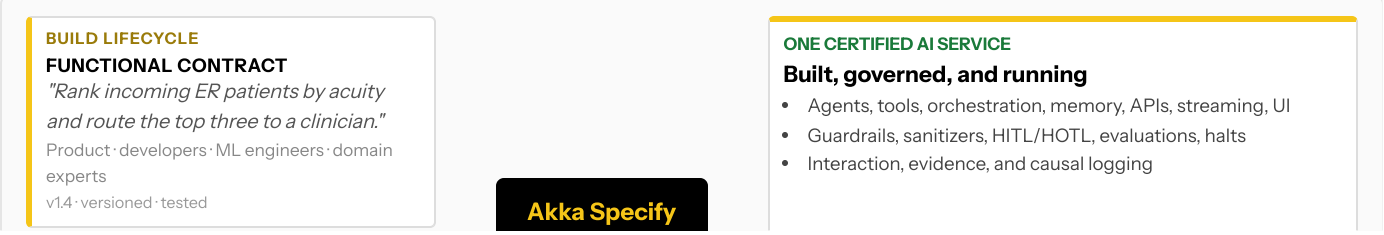
High-risk AI carries a 10-year logging-retention obligation (Art. 72), enforceable since Feb 2025 (prohibited) / Aug 2025 (high-risk).

How Akka governs

At the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; atomic PII scrub-with-explain; pre-deployment classification against **241 regulations and frameworks and 1,718 controls — 699 of which carry a financial penalty**; a multi-persona sign-off recipe engine; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. The governance lifecycle is versioned and tested independently of the build.

Two Lifecycles, One Certified System

Building on AgentCore means engineers wiring and operating a set of AWS services. Security and compliance teams can write AgentCore Policy rules in natural language without changing agent code, and those rules cover tool calls through an AgentCore Gateway. Akka runs two independent lifecycles on one platform via **Akka Specify**:





The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences.

Real-Time Streaming at Petabyte Scale

Streaming is not a native AgentCore service; real-time pipelines are provisioned and operated separately on other AWS services. Akka streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

For the Buyer: Maturity, Lock-In, and Accountability

Buyer concern	AWS Bedrock AgentCore	Akka
Product maturity	GA October 13, 2025; core capabilities still landing (Policy and Evaluations GA March 2026; Guardrails in Policy GA June 2026)	since 2009, 100,000+ deployments, 52 banks
What you operate	AgentCore services + model access + Guardrails + Knowledge Bases, integrated and run by the customer	One platform, one SLA, 24/7 SRE — Akka owns the running system
Portability / lock-in	AWS-only: IAM service-linked roles, VPC/PrivateLink, CloudFormation, CloudWatch, proprietary service APIs	Any cloud, own Kubernetes, on-prem, sovereign cloud; portable specs
Certifications & audits	Bedrock: ISO, SOC, CSA STAR L2, GDPR, FedRAMP High, HIPAA-eligible	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Risk transfer	Service credits on the model-API SLA	Availability and data-integrity guarantees backed by contractual indemnities
Budget predictability	Consumption meters across the AgentCore services plus model spend, scaling with load	Fixed annual fee finance can forecast

The structural point is not whether AWS will be around — it will. It is that AgentCore is a recent, consumption-metered collection of services the customer integrates and runs inside AWS, while Akka is an integrated, governed, portable platform with the SLA owned by the vendor.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
2,000
developers across 100 projects on one governed platform

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We're an AWS shop already. Why not just use AgentCore?
You can, but you are assembling and operating AgentCore services plus Bedrock model access, Guardrails, and Knowledge Bases — each billed and run separately, all locked to AWS. Akka delivers the same capabilities as one integrated runtime with a six-nines SLA Akka owns, and it runs in your AWS VPC, on another cloud, on your own

Can't we add EU AI Act compliance with Guardrails and Policy?
Policy blocks tool calls that break a rule and can require human approval, and Guardrails screen those calls for prompt injection and sensitive data. Neither maps to a named regulation. The EU AI Act expects classification before deployment, immutable records witnessed as decisions happen, human override of running processes,

Kubernetes, or on-prem. You get the platform instead of the integration project.

AgentCore is from AWS — isn't it the safe, mature choice?
AWS is durable; AgentCore is recent. It reached general availability in October 2025, and core governance and evaluation capabilities were still reaching GA into 2026 (Policy and Evaluations March 2026; Guardrails in Policy June 2026). Akka has operated since 2009 across 100,000+ production deployments behind a single, integrated platform.

and a sealed audit artifact. Akka embeds classification against 241 regulations and frameworks and 1,718 controls, hash-chained evidence, HITL/HOTL control, and pre-deployment governance inline.

Isn't AgentCore's consumption pricing cheaper than a fixed fee?
AgentCore meters its services on per-vCPU-hour, per-request, and per-token units, on top of model token spend, and the bill scales with load. The Akka shared-compute model is up to 90% cheaper to operate for the same agentic transaction volume, on a fixed annual fee finance can forecast.

Sources

AgentCore GA & components: aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available — GA October 13, 2025; seven modular services (Runtime, Memory, Gateway, Identity, Code Interpreter, Browser, Observability); VPC, PrivateLink, CloudFormation at GA
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AgentCore governance: aws.amazon.com/bedrock/guardrails (model-layer content/PII safety); Policy = tool-access-layer enforcement; aws.amazon.com/about-aws/whats-new/2026/08/temporal-policies-agentcore (temporal policies, human approval); docs.aws.amazon.com/audit-manager/latest/userguide/aws-generative-ai-best-practices.html (72 automated + 38 manual controls); aws.amazon.com/about-aws/whats-new/2026/06/aws-security-hub-cspm-ai-security (31 controls)
AWS lock-in / VPC / IAM: docs.aws.amazon.com/bedrock-agentcore/latest/devguide/vpc-interface-endpoints.html; .../security-iam-awsmanpol.html — IAM service-linked roles, VPC/PrivateLink, CloudWatch-backed observability
AgentCore pricing: aws.amazon.com/bedrock/agentcore — consumption-based across the AgentCore services (per-vCPU-hour + per-GB-hour runtime, plus memory, gateway, tools, observability); Bedrock model inference billed separately; exact rates per the AWS pricing page
Bedrock compliance: aws.amazon.com/bedrock/security-privacy-responsible-ai — ISO, SOC, CSA STAR L2, GDPR, FedRAMP High, HIPAA-eligible
Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies
Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python
Akka platform: 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); 241 regulations and frameworks / 1,718 controls / 699 with a financial penalty; 100,000+ deployments / since 2009; profitable; Dell Technologies Capital